
Table of Contents

.....	1
PARAMETERS	1
TRANSMITTER	1
ADD CYCLIC PREFIX	1
CHANNEL	2
RECEIVER	2
BER	2
PLOT 1: CYCLIC PREFIX	2
PLOT 2: TRANSMITTED CONSTELLATION	3
PLOT 3: RECEIVED CONSTELLATION	4

```
clc;  
clear;  
close all;
```

PARAMETERS

```
M = 16;  
k = log2(M);
```

```
Nfft = 64;  
Nofdm = 100;
```

```
CP = 16;
```

TRANSMITTER

```
Nbits = Nofdm * Nfft * k;
```

```
% Generate random bits  
txBits = randi([0 1], Nbits, 1);
```

```
% 16-QAM modulation  
txSymbols = qammod(txBits, M, ...  
    'InputType', 'bit', ...  
    'UnitAveragePower', true);
```

```
% Arrange symbols into OFDM symbols  
txSymbolsMatrix = reshape(txSymbols, Nfft, Nofdm);
```

```
% IFFT  
txOFDM = ifft(txSymbolsMatrix, Nfft);
```

ADD CYCLIC PREFIX

```
% Take last CP samples  
cyclicPrefix = txOFDM(end-CP+1:end, :);
```

```
% Add CP
txOFDM_CP = [cyclicPrefix; txOFDM];
```

CHANNEL

```
% No noise and no multipath yet
rxOFDM_CP = txOFDM_CP;
```

RECEIVER

```
% Remove cyclic prefix
rxOFDM = rxOFDM_CP(CP+1:end, :);

% FFT
rxSymbolsMatrix = fft(rxOFDM, Nfft);

% Convert to serial
rxSymbols = rxSymbolsMatrix(:);

% 16-QAM demodulation
rxBits = qamdemod(rxSymbols, M, ...
    'OutputType', 'bit', ...
    'UnitAveragePower', true);
```

BER

```
[numErrors, ber] = biterr(txBits, rxBits);

fprintf('Number of bit errors = %d\n', numErrors);
fprintf('BER = %e\n', ber);

Number of bit errors = 0
BER = 0.000000e+00
```

PLOT 1: CYCLIC PREFIX

```
figure;

plot(real(txOFDM(:,1)), 'LineWidth', 1.2);

hold on;

plot(real(txOFDM_CP(:,1)), 'LineWidth', 1.2);

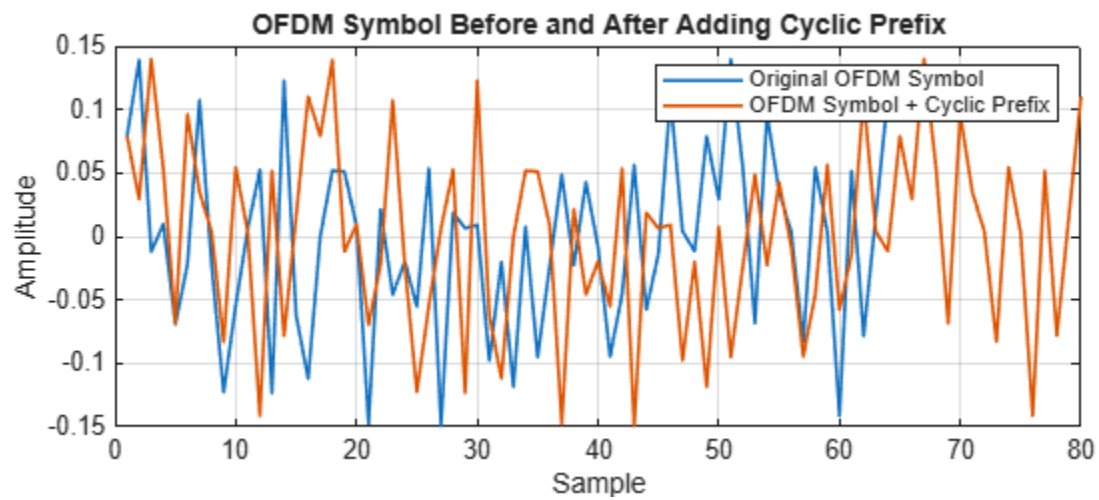
xlabel('Sample');
ylabel('Amplitude');

title('OFDM Symbol Before and After Adding Cyclic Prefix');

legend('Original OFDM Symbol', ...
```

```
'OFDM Symbol + Cyclic Prefix');
```

```
grid on;
```



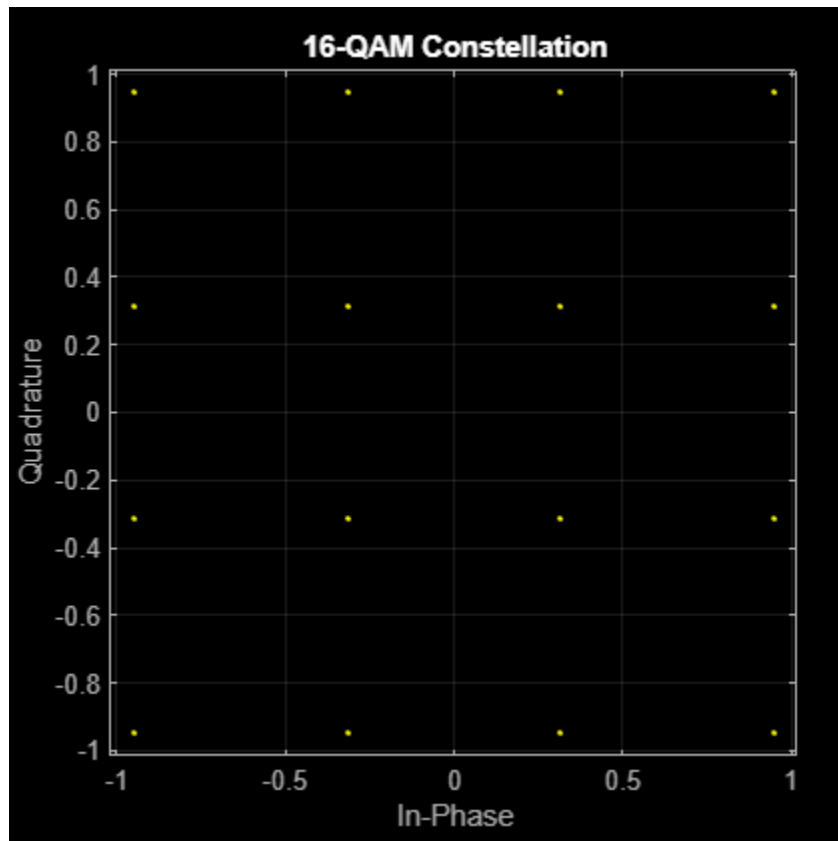
PLOT 2: TRANSMITTED CONSTELLATION

```
figure;
```

```
scatterplot(txSymbols);
```

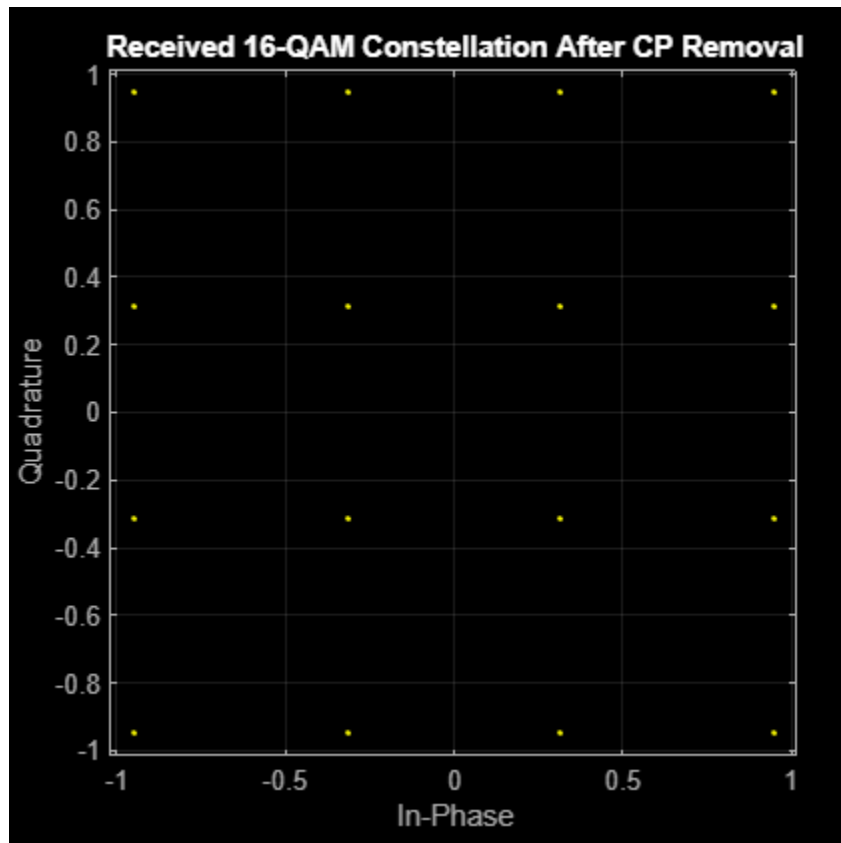
```
title('16-QAM Constellation');
```

```
grid on;
```



PLOT 3: RECEIVED CONSTELLATION

```
figure;  
scatterplot(rxSymbols);  
title('Received 16-QAM Constellation After CP Removal');  
grid on;
```



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